

The Iterated Prisoner's Dilemma

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1 Introduction

The Prisoner's Dilemma is one of the most well-known problems in game theory. It describes a situation in which players must independently choose between cooperation and defection. Although cooperation can lead to a better outcome for both players, each player has an individual incentive to defect. This creates a conflict between individual incentives and the outcome that is collectively beneficial.

The Prisoner's Dilemma has been studied in many different forms, particularly when the game is played repeatedly. This paper will examine how different conditions of the game affect the outcomes observed in existing research. In particular, it will consider how factors such as the number of rounds, payoff structure, number of players, and strategies being used can influence the behaviour and performance observed in the game.

2 Background

2.1 The Prisoner's Dilemma

The Prisoner's Dilemma is a game in which players independently choose between two possible actions: cooperation and defection. The payoff received by each player depends on both their own action and the action chosen by the other player.

For a two-player Prisoner's Dilemma, the payoffs can be represented by the following matrix:

	C	D
C	(R, R)	(S, T)
D	(T, S)	(P, P)

where C represents cooperation and D represents defection. The four payoff values are usually required to satisfy

$$T > R > P > S.$$

Here, T is the temptation payoff received by a player who defects while the other player cooperates, R is the reward received when both players cooperate, P is the punishment received when both players defect, and S is the sucker's payoff received by a player who cooperates while the other player defects.

One commonly used set of payoffs is

$$T = 5, \quad R = 3, \quad P = 1, \quad S = 0.$$

Under these conditions, defection is a dominant strategy. Regardless of the action chosen by the other player, a player receives a higher payoff by defecting.

Therefore, when the game is played only once, both players choosing to defect forms a Nash equilibrium. However, mutual cooperation would give both players a higher payoff than mutual defection. This conflict between individual incentives and the collectively better outcome is the central dilemma of the game.

2.2 The Iterated Prisoner's Dilemma

The Prisoner's Dilemma can also be played repeatedly between the same players. This is known as the iterated Prisoner's Dilemma.

Instead of making a single decision, players make decisions over a sequence of rounds. The outcome of one round can therefore provide information that can be used when making a decision in a later round. This introduces the possibility that a player's current behaviour can depend on the previous behaviour of their opponent.

For example, a player may cooperate in an initial round and then continue cooperating if their opponent also cooperates. Alternatively, they may respond to a previous defection by defecting in the next round.

The repeated nature of the game therefore introduces strategies that are not relevant in the one-shot version. A strategy in the iterated Prisoner's Dilemma can be viewed as a rule that determines a player's action based on the information available from previous rounds.

2.3 Strategies in the Iterated Prisoner's Dilemma

Different strategies can produce very different behaviour when repeatedly interacting with one another.

Some simple examples are:

- **Always Cooperate:** the player cooperates in every round.
- **Always Defect:** the player defects in every round.
- **Tit-for-Tat:** the player cooperates in the first round and then copies the opponent's action from the previous round.

These strategies illustrate why the iterated version of the Prisoner's Dilemma is different from the one-shot version. For example, two players using Always Defect will defect in every round, while two players using Tit-for-Tat will cooperate in every round if they both begin by cooperating.

When different strategies interact, however, the sequence of outcomes can become more complicated. A strategy that performs well against one opponent may not perform equally well against another.

2.4 Why the Iterated Prisoner's Dilemma Is Interesting

The iterated Prisoner's Dilemma provides a way to study how behaviour changes when players interact repeatedly rather than only once.

One important area of interest is cooperation. Although defection is the dominant strategy in the one-shot game, repeated interaction can allow cooperation to persist through strategies that respond to previous behaviour.

Another important aspect is the interaction between strategies. The outcome of a game does not depend only on the strategy used by one player. It can also depend on the strategy used by the opponent. Consequently, the performance of a particular strategy may change when it is placed in a different strategic environment.

These observations raise questions about whether conclusions about the Prisoner's Dilemma are consistent across different ways of setting up the game.

2.5 Conditions Studied in Previous Research

Researchers have studied the Prisoner's Dilemma under a wide range of conditions. Rather than treating the game as having one fixed setup, different studies have changed various aspects of the game to investigate how these changes affect its outcomes.

Some of the conditions that have been varied include:

- **Number of rounds:** outcomes may change based on the number of rounds that are played.
- **Payoff structure:** the values of T , R , P , and S can be changed, altering the incentives associated with cooperation and defection.
- **Number of players:** although the traditional Prisoner's Dilemma is a two-player game, researchers have also considered versions involving more players.
- **Strategies:** studies can compare different sets of strategies, which can affect the relative performance of the strategies being investigated.
- **Noise and errors:** some studies allow players to make errors, meaning that the action actually played may differ from the action intended by the player.
- **Ending conditions:** repeated games can differ in whether the number of rounds is known in advance or whether there is uncertainty about when the game will end.

The following sections examine research involving these different conditions and compare the effects that they have on the outcomes of the Prisoner's Dilemma.

3 Effect of Game Conditions

3.1 Number of Rounds

The number of rounds can affect the way players behave in the Iterated Prisoner’s Dilemma because repeated interaction gives players more opportunities to respond to their opponent’s previous actions. This also means that players may behave differently depending on how close they are to the end of the game. Two studies that help illustrate this are Wang et al. (2016) and Mao et al. (2017).

Wang et al. (2016) directly investigated the effect of game length by having participants play games consisting of either 60 or 500 rounds. Participants played against computer-controlled strategies, including an *extortionate* strategy and a *generous* strategy. An extortionate strategy is one that accepts a lower payoff for itself in order to obtain a larger payoff relative to its opponent, while a generous strategy is more willing to cooperate even when this does not maximise its own payoff. The researchers also varied whether participants knew that their opponent was a computer.

The difference between 60 and 500 rounds was important because the longer games gave players considerably more opportunity to learn and adapt to their opponent. For example, among participants who knew they were playing a computer, the cooperation rate during the first 100 rounds of the 500-round games was approximately 0.563, while it increased to approximately 0.757 during the final 100 rounds. A Wilcoxon signed-rank test found this increase to be statistically significant ($p < 0.001$). The researchers also found that the extortionate strategy performed better relative to the generous strategy in the 500-round condition, an advantage that was not observed in the 60-round condition.

These results suggest that longer games can change the development of cooperation, but they do not mean that simply increasing the number of rounds always produces more cooperation. The effect also depends on factors such as what strategy the opponent uses and whether the player knows what they are playing against. In particular, the 500-round games gave participants much more time to learn the behaviour of their opponent, making it difficult to separate the effect of game length itself from the effect of having more opportunities for adaptation.

Mao et al. (2017) examined a different aspect of repeated interaction. Instead of comparing different game lengths, they kept each game at 10 rounds and looked at how behaviour changed both within a game and across many repeated games. Their experiment involved 94 participants who played 20 ten-round games per day for 20 consecutive weekdays, with participants being randomly assigned new partners after each game. This allowed the researchers to observe how cooperation developed as participants gained experience.

They found that, during the first several days, defection tended to move towards earlier rounds of the ten-round game. This process, known as *unravelling*, means that players who initially defected only near the end of the game began defecting earlier. However, this process eventually stabilised after approximately one week. Around 40% of the participants were also identified as “resilient cooperators”, who continued to cooperate even when other players were increasingly defecting.

Together, these studies suggest that the number of rounds is important, but its effect is not simply a matter of “more rounds means more cooperation” or “more rounds means less

cooperation.” Wang et al. show that changing the length of the game can affect cooperation and the relative performance of strategies, while Mao et al. show that behaviour can also change depending on where a player is within a finite game and how much experience they have accumulated.

Therefore, the evidence suggests that the length of a repeated game can influence how cooperation develops, but the relationship is not necessarily straightforward. Longer games provide more opportunities for players to learn and adapt, while behaviour can also depend on the position of a round within the game and experience from previous interactions.

3.2 Payoff Structure

The payoff structure of the Prisoner’s Dilemma determines how attractive cooperation and defection are to the players. The four payoffs are usually denoted by T , R , P , and S , where T is the payoff for defecting when the other player cooperates, R is the payoff when both players cooperate, P is the payoff when both defect, and S is the payoff for cooperating when the other player defects. For the game to be a Prisoner’s Dilemma, these payoffs satisfy

$$T > R > P > S.$$

Therefore, a player always has a higher individual payoff from defecting than cooperating, regardless of what the other player does. However, mutual cooperation gives both players more than mutual defection because $R > P$.

Gächter et al. (2024) investigated whether changing these payoff values affects people’s willingness to cooperate. They conducted three experiments involving a total of 1,993 participants and varied the payoffs across different one-shot Prisoner’s Dilemma games. Their main experiments were designed so that different aspects of the payoff structure could be varied separately, allowing them to investigate which features of the payoffs were related to cooperation.

One of the main quantities they considered was the efficiency gain from cooperation, which measures how much better mutual cooperation is compared with mutual defection. In other words, it compares the payoff R from both players cooperating with the payoff P from both players defecting. They also considered two other features: the temptation to defect against a cooperating player and the risk of cooperating when the other player defects.

The researchers found a robust relationship between the efficiency gain and cooperation. When the gain from mutual cooperation over mutual defection was larger, participants were more likely to cooperate. In their main analysis, a 0.1 increase in their efficiency measure was associated with a 3.99 percentage-point increase in the probability of cooperation. In contrast, the effect of the risk of being exploited was not statistically significant, while the effect of the temptation to defect was less consistent across their experiments.

This suggests that changing the payoff structure does not affect cooperation simply because one particular payoff becomes larger. Instead, what matters is the relationship between the different payoffs and, in particular, how attractive mutual cooperation is compared with mutual defection. Therefore, the payoff structure can influence cooperation even when the game remains a Prisoner’s Dilemma.

References

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